

2. Loss of suitable habitat by destruction, degradation or fragmentation is the most important cause of species becoming endangered around the world.

(a) (i) Distinguish between the terms endangered and extinct. [1]

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(ii) Apart from habitat loss, name **two** other causes of species extinction. [2]

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Following a forest fire, given time, the forest will regenerate.

(b) (i) Give the full name of the biological process involved in forest regeneration. [1]

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(ii) Using relevant **technical terms**, describe and explain the process of forest regeneration. [3]

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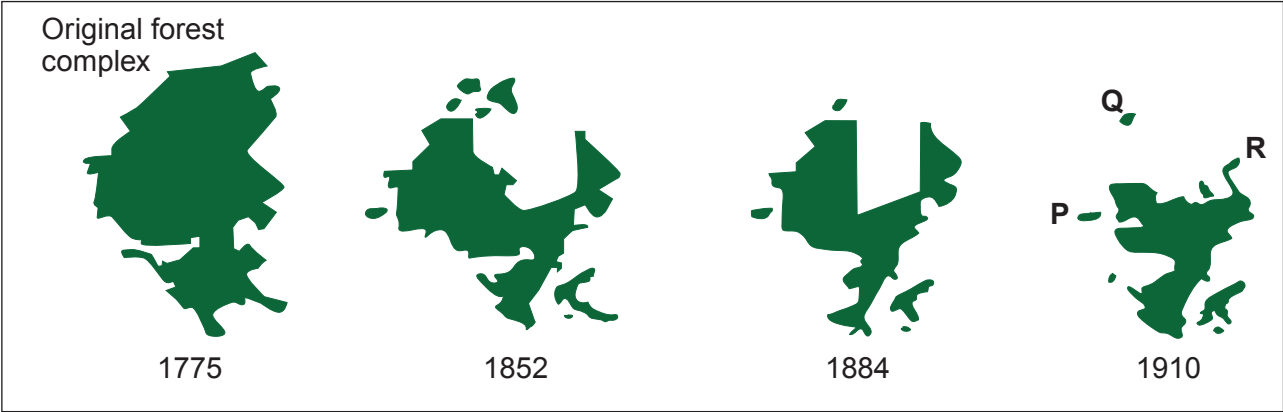
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- (c) Human activity has resulted in the forest landscapes of Europe existing as a mixture of fragments of different ages and sizes. **Image 2.1** shows the fragmentation of forests surrounded by developed land. Three fragments of forest **P**, **Q** and **R** are labelled on the 1910 map.

Image 2.1



- (i) The Hardy-Weinberg principle states that allele frequencies remain constant from generation to generation. Explain **two** reasons why this would not apply to the gene pools of species in fragments **P**, **Q** and **R**. [2]

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- (ii) List fragments **P**, **Q** and **R** in order of risk of species extinction, from most to least. Explain your answer. [4]

Most risk Least risk

Explanation

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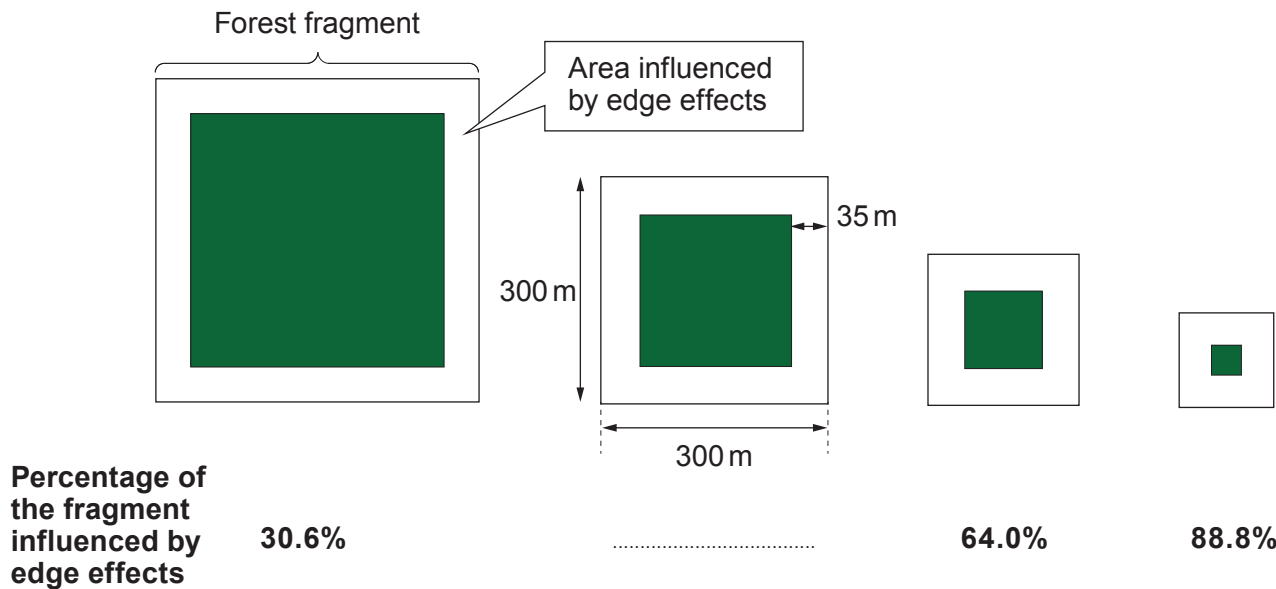
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- (d) As well as gene pool effects, fragmentation exposes species to potentially harmful environmental effects. These are called 'edge effects' because conditions at the edge of a forest differ from those further inside. **Image 2.2** shows how square forest fragments of different sizes are affected by edge effects.

Image 2.2



- (i) Assuming edge effects extend 35 m into each habitat patch, calculate the percentage of the fragment influenced by the edge effects for the second largest patch.
Write your answer in the space in Image 2.2. [2]
Space for working.

- (ii) Describe the trend in the data and suggest **two** abiotic conditions that would differ in the area affected by edge effects. [3]

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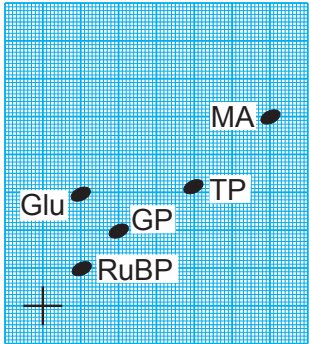
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Experiment 2 was carried out using exactly the same method as **Experiment 1**, except that the mineral solution used contained no nitrate ions. The results are shown in **Image 5.3**.

Image 5.3

Experiment 2. Mineral solution lacking nitrate.



15 seconds

- (ii) Interpret the results of **Experiment 2** to explain the difference between the 15-second autoradiograms from **Experiments 1** and **2** and suggest the nature of compound **X** shown on the **15-second** autoradiogram in **Experiment 1**. [2]

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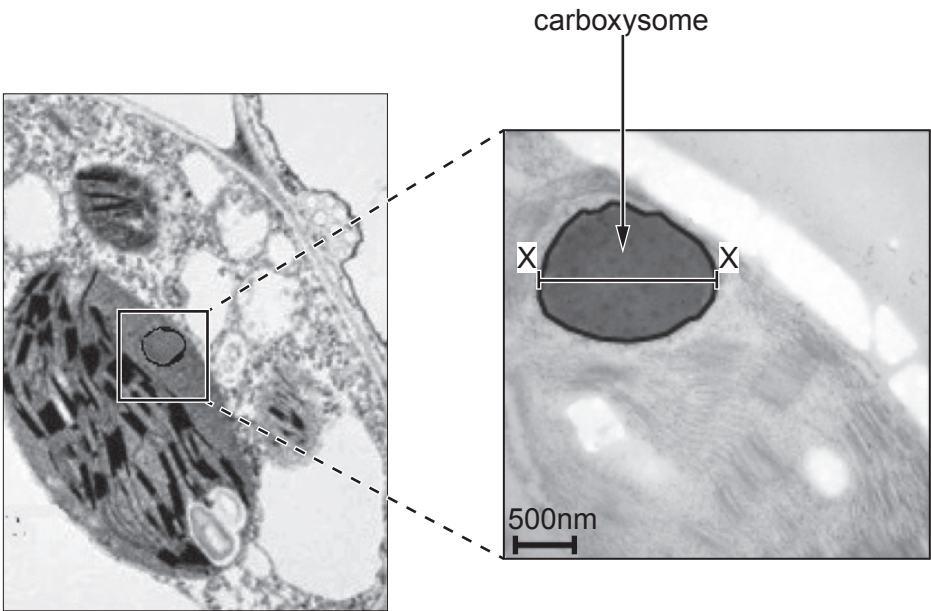
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In Cyanobacteria the carbon fixing reaction, catalysed by RuBisCO, is carried out in micro-compartments called carboxysomes. Scientists have used genetic engineering techniques to produce maize plant cells which have carboxysomes. **Image 5.4** shows part of such an engineered cell.

Image 5.4



- (c) (i) With reference to **Image 5.4**, describe precisely where the carboxysome is located. [1]

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- (ii) Use the scale bar to calculate the length, X–X, of the carboxysome in μm . [2]

Length = μm



Examiner
only

In addition to RuBisCO, carboxysomes contain the enzyme carbonic anhydrase. Having carbonic anhydrase inside **enhances uptake of carbon dioxide** by catalysing its combination with water.

- (d) (i) Suggest an advantage to farmers of introducing carboxysomes into crop plants. Explain your answer. [2]

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- (ii) Predict the effect that widespread use of crops engineered to contain carboxysomes might have on the carbon cycle and explain the impact this effect might have on the prospects for endangered species. [3]

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